Participant Information and Consent

Multilingual Fairness Study in LLM-Supported Generative AI Learning

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1. Researchers and Contact Information

If you have questions about this study, your rights, or the handling of your data, please contact:

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This study is conducted as part of a Master's thesis at FH Dortmund in collaboration with KU Leuven.

2. Purpose of the Study

This study investigates how the language used to interact with a Large Language Model (LLM) influences learning outcomes and user experience in a short “Introduction to Cancer Biology” course.

All participants receive the same learning materials. These materials consist of English-language slides and an English transcript based on the same Cancer Biology content. The experimental condition is the interaction language with the LLM. The study includes five language conditions: English, Dutch, Turkish, Hindi, and Albanian (optionally German).

The AI learning assistant is powered by Google Gemini, specifically the Gemini 2.5 Flash model. Responses are instructed and checked to remain in the assigned language throughout the study.

3. What Participation Involves

If you agree to participate, you will complete the following steps:

A short background survey about language skills, Cancer Biology background and interest, AI experience, and basic demographics.

A guided learning session using the platform and the AI tutor. You may browse 26 Cancer Biology slides, request slide-based explanations, and ask questions in your assigned language. The learning materials themselves remain in English.

A short Cancer Biology knowledge test consisting of 5 multiple-choice questions.

A user experience questionnaire (UEQ) with 26 standard items and 1 mandatory free-text feedback question. The free-text response requires a minimum length. For participants in non-English conditions, part of the questionnaire asks for a comparison between the assigned-language experience and the participant's imagined experience of using the AI tutor in English.

Navigation through the platform is one-way only. After moving to the next section, you cannot return to previous sections. This is done to protect the integrity of the study procedure.

The total duration is approximately 60-90 minutes, depending on your pace.

4. Voluntary Participation and Withdrawal

Participation is voluntary. You may stop at any time without giving a reason and without negative consequences. If you withdraw, the researcher will exclude your data from analysis where feasible, depending on the stage of the study and the degree of anonymization.

5. What Data Will Be Collected

During the study, we may collect and process the following data:

5.1 Background survey (profile)

- Native language (based on your assigned language credential).
- Self-rated proficiency in English and in your native language.
- Familiarity with generative AI tools and prior usage frequency.
- Self-rated prior knowledge of Generative AI concepts.
- Whether you have taken a course or formal training in AI or machine learning.
- Age, gender, current level of study, and field of study.
- Preferred language when learning new course content.

5.2 Learning session data

- A pseudonymous session ID assigned by the system.
- The assigned language condition.
- Your questions/prompts to the AI tutor and the AI tutor's responses.
- Interaction counts (e.g., slide explanations vs. manual chat) and timestamps.
- Technical metadata (page durations, error codes, and automated language-detection checks).

5.3 Tests and questionnaires

- Your answers and score on the Cancer Biology knowledge test.
- Your responses to the UEQ.
- Your mandatory free-text feedback response
- For participants in non-English language conditions, comparative questionnaire information about the assigned-language experience in relation to the participant's experience of using the AI tutor in English.

5.4 Technical presence information

A pseudonymous identifier, current page, and time of last activity are processed for concurrency management and for detecting abandoned sessions.

The platform uses a browser heartbeat mechanism that sends a technical presence signal approximately every 20 seconds. If no heartbeat is received for 2 minutes, the session may be marked as abandoned. The heartbeat contains only technical presence information linked to the pseudonymous session ID.

To protect the study procedure and ensure fair access to the AI service, the system also limits the number of active concurrent sessions.

The study does not require sensitive personal information (for example, health data, political opinions, or religious beliefs). Please do not enter such information into the chat.

6. Data Storage, Security, and Retention

Your data are stored under a pseudonymous code (session ID). Your real name is not stored together with the study data. Access to the data is restricted to the research team.

Data may be stored on secure university systems and on a secure cloud service (Supabase) with restricted access for research data management.

Access is restricted to the research team, namely the student researcher and the supervisor. Data will be kept for up to 5 years after the end of the study, and then deleted or further anonymized.

7. Data Protection and Your Rights (GDPR)

The processing of personal data is carried out for scientific research in higher education. Depending on institutional policy, the legal basis may be a task in the public interest (GDPR Article 6(1)(e)) or legitimate interests (GDPR Article 6(1)(f)).

You have rights under the GDPR, including the right to:

- request information about how your data are processed;
- access your data and request correction where appropriate;
- request restriction of processing or object to processing in certain cases.

Because this is scientific research, some rights may be limited if applying them would make the research impossible or seriously impair it (for example, if the data have already been fully pseudonymized and analysed).

8. Risks and Benefits

This study involves minimal risk. Possible inconveniences include time investment and mild stress during the learning session or test.

Potential benefits include learning about Cancer Biology and contributing to research on fairer multilingual AI-supported learning and AI-assisted science education.

9. Reuse of Pseudonymized Data

Pseudonymized data may be reused for related scientific research on AI-assisted science education, Cancer Biology learning, AI in education, or multilingual learning, and for teaching purposes using anonymized examples. Any reuse will follow data-protection law and institutional policies.

10. Consent

Before proceeding to the Profile Survey, you must select the consent checkbox stating: "I have read the documents above and give my consent to participate."

By continuing in the platform and selecting the consent checkbox, you confirm that:

- You have read and understood this document and had the opportunity to ask questions.
- You understand what participation involves and that participation is voluntary.
- You understand which data will be collected, how they will be stored, and how they will be used for research.
- You understand that your data are processed under a pseudonymous session ID and you will not be personally identifiable in publications.
- You understand that you can withdraw at any time without negative consequences.